STEVENS INSTITUTE OF TECHNOLOGY

Data Acquisition and Processing I: Big Data

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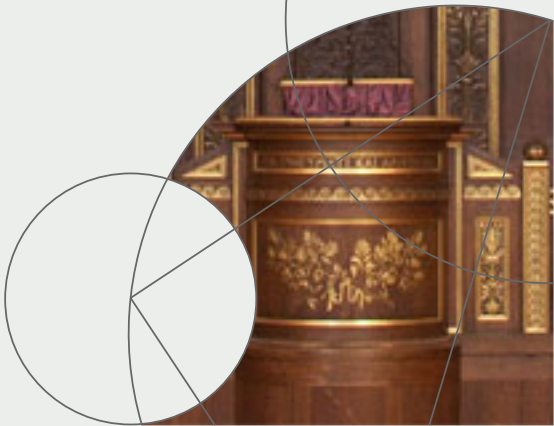


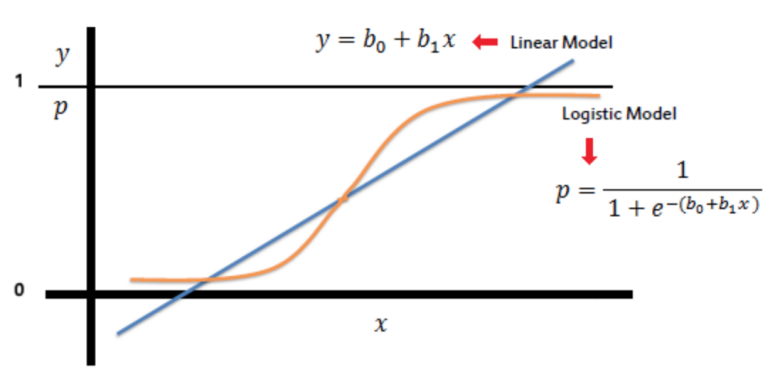
Logistic Regression vs. Linear Regression

- Linear Regression : Regression

$$\mathbf{y} = b_0 + b_1\mathbf{x}_1 + b_2\mathbf{x}_2 + \cdots + b_n\mathbf{x}_n$$

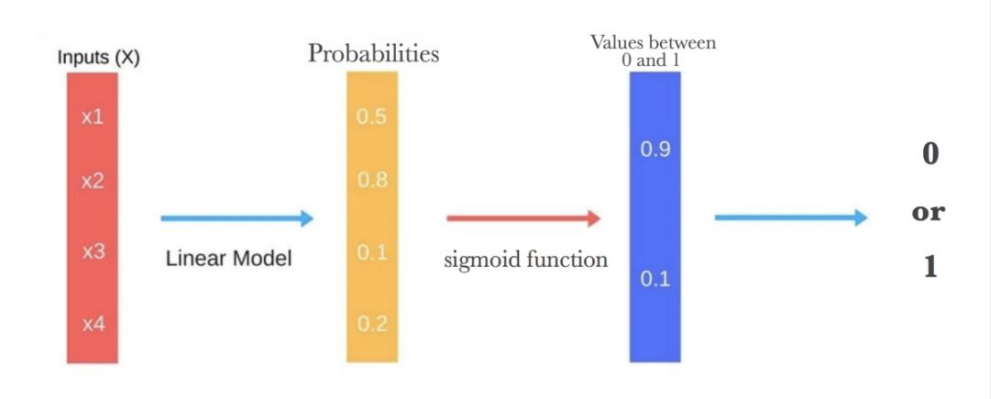
- Logistic regression : Classification

$$\text{Prob.}(\mathbf{y} = 1) = \frac{1}{1 + \exp(-[b_0 + b_1\mathbf{x}_1 + b_2\mathbf{x}_2 + \cdots + b_n\mathbf{x}_n])}$$



Logistic Regression vs. Linear Regression

- Logistic regression



Logistic Regression vs. Linear Regression

- Logistic regression output as probabilities
- Logistic regression target variable is categorical
- Logistic regression use the logistic function to transform the linear mapping variables

Logistic Regression Interpretation

- The interpretation of the weights in logistic regression differs from the interpretation of the weights in linear regression, since the outcome in logistic regression is a probability between 0 and 1.
- The weights do not influence the probability linearly any longer. The weighted sum is transformed by the logistic function to a probability. Therefore we need to reformulate the equation for the interpretation so that only the linear term is on the right side of the formula.

$$\log\left(\frac{P(\mathbf{y}=1)}{1-P(\mathbf{y}=1)}\right)=\log\left(\frac{P(\mathbf{y}=1)}{P(\mathbf{y}=0)}\right)=b_0+b_1\mathbf{x}_1+b_2\mathbf{x}_2+\cdots+b_n\mathbf{x}_n$$

- We call the term in the log() function "odds" (probability of event divided by probability of no event) and wrapped in the logarithm it is called log odds.
- This formula shows that the logistic regression model is a linear model for the log odds.

ROC for Logistic Regression

- We use the receiver operating characteristic curve, or ROC, to evaluate the logistic regression performance.
- We first put the ground truth of the two classes "1" and "0" in the table in parallel with the logistic regression prediction probablities.

Truth	prob.
1	0.9
1	0.7
1	0.6
1	0.4
0	0.7
0	0.5
0	0.4
0	0.2
0	0.2

ROC for Logistic Regression

- We use the receiver operating characteristic curve, or ROC, to evaluate the logistic regression performance.
- We first put the ground truth of the two classes "1" and "0" in the table in parallel with the logistic regression prediction probablities.
- We then put the histograms of these two classes.

Truth	prob.
1	0.9
1	0.7
1	0.6
1	0.4
0	0.7
0	0.5
0	0.4
0	0.2
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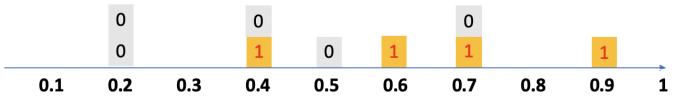
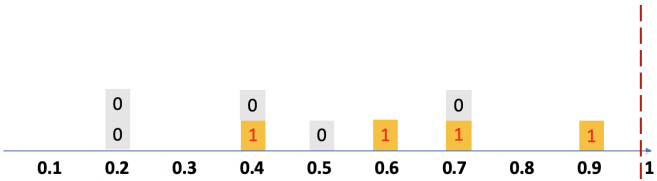


ROC for Logistic Regression

- We choose a threshold value over the histograms of these two classes (starting from the largest value 1.0). Then, $P \geq \text{Threshold} \Rightarrow 1$ vs. $P < \text{Threshold} \Rightarrow 0$.
- You will have
  - False positive: prediction "1" with selected threshold but the ground truth is "0"
  - True positive: prediction "1" with selected threshold but the ground truth is "1"

Truth	prob.
1	0.9
1	0.7
1	0.6
1	0.4
0	0.7
0	0.5
0	0.4
0	0.2
0	0.2



Threshold	1	0.9	0.7	0.6	0.5	0.4	0.2
False Positive	0						
True Positive	0						

ROC for Logistic Regression

- We keep change different threshold value over the histograms of these two classes (starting from the largest value 1.0).

Truth	prob.
1	0.9
1	0.7
1	0.6
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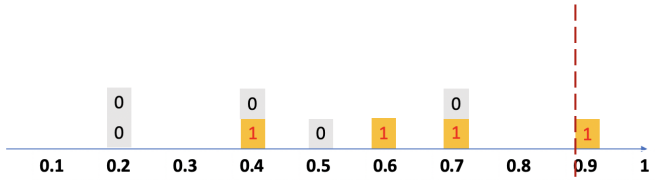

Threshold	1	0.9	0.7	0.6	0.5	0.4	0.2
False Positive	0	0					
True Positive	0	1/4					

ROC for Logistic Regression

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1	0.9
1	0.7
1	0.6
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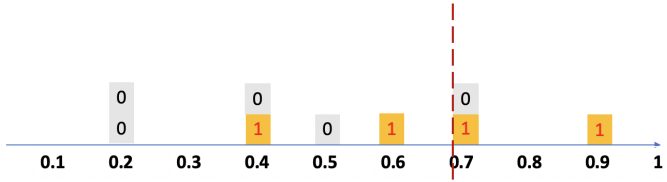

Threshold	1	0.9	0.7	0.6	0.5	0.4	0.2
False Positive	0	0	1/5				
True Positive	0	1/4	2/4				

ROC for Logistic Regression

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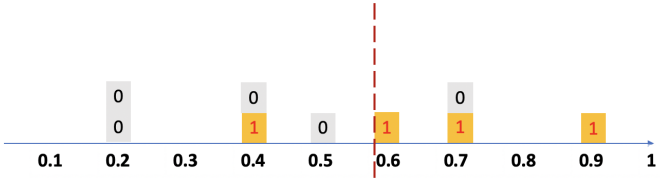
0	0.2



Threshold	1	0.9	0.7	0.6	0.5	0.4	0.2
False Positive	0	0	1/5	1/5			
True Positive	0	1/4	2/4	3/4			

ROC for Logistic Regression

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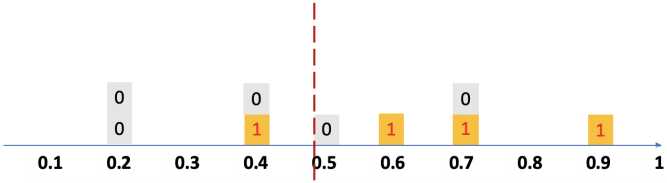

Threshold	1	0.9	0.7	0.6	0.5	0.4	0.2
False Positive	0	0	1/5	1/5	2/5		
True Positive	0	1/4	2/4	3/4	3/4		

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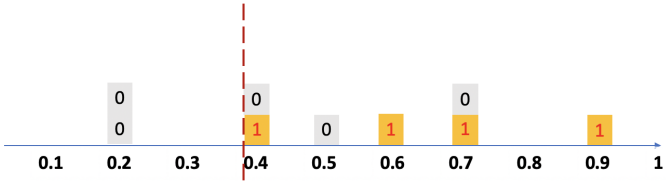
0	0.2



Threshold	1	0.9	0.7	0.6	0.5	0.4	0.2
False Positive	0	0	1/5	1/5	2/5	3/5	
True Positive	0	1/4	2/4	3/4	3/4	1	

ROC for Logistic Regression

- We keep change different threshold value over the histograms of these two classes (starting from the largest value 1.0).

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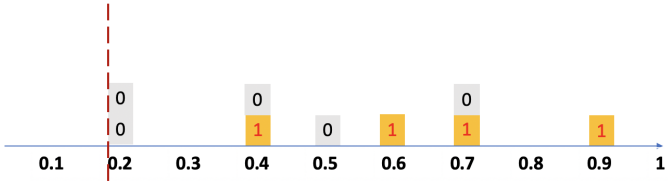
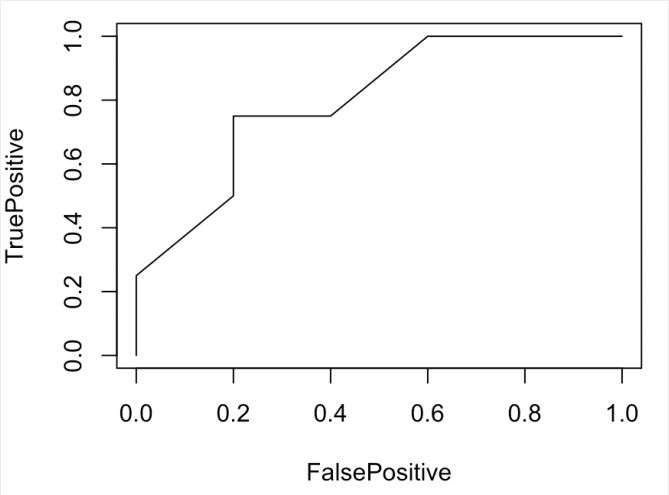


Threshold	1	0.9	0.7	0.6	0.5	0.4	0.2
False Positive	0	0	1/5	1/5	2/5	3/5	1
True Positive	0	1/4	2/4	3/4	3/4	1	1

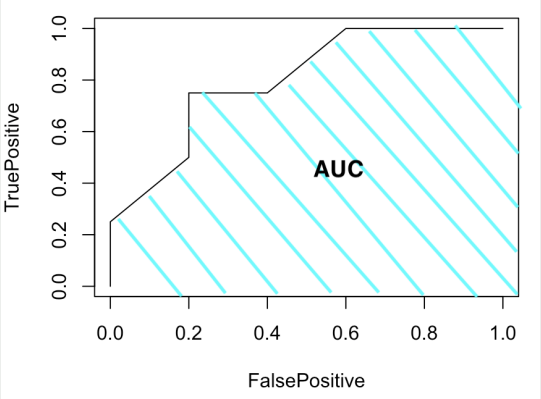
ROC for Logistic Regression

- We finally plot the true positive vs. false positive to obtain the ROC curve.



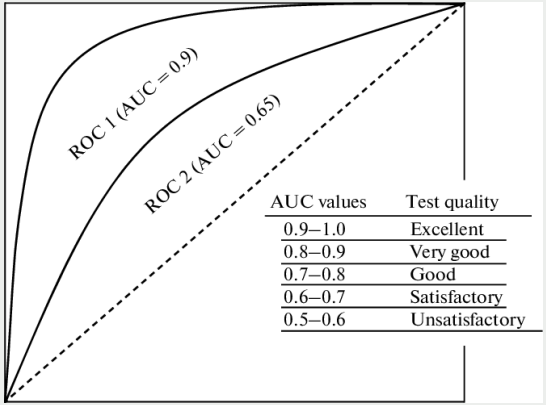
ROC vs. AUC

- In order to compare different models in terms of ROC curve, we use the Area Under Curve (AUC) to evaluate the model performance.
- The ideal model will be a lower false positive rate with a higher true positive value. In other word, to obtain a better, we need to push the ROC to the left upper corner. Then the AUC will be larger.



AUC

- Different models can be evaluated in ROC and compared by the associated AUC values.



AUC values	Test quality
0.9–1.0	Excellent
0.8–0.9	Very good
0.7–0.8	Good
0.6–0.7	Satisfactory
0.5–0.6	Unsatisfactory